

EXPERIMENT 4: VACUUM CLEANER PROBLEM

Aim

To write a Python program to solve the **Vacuum Cleaner Problem** using Artificial Intelligence techniques.

Objective

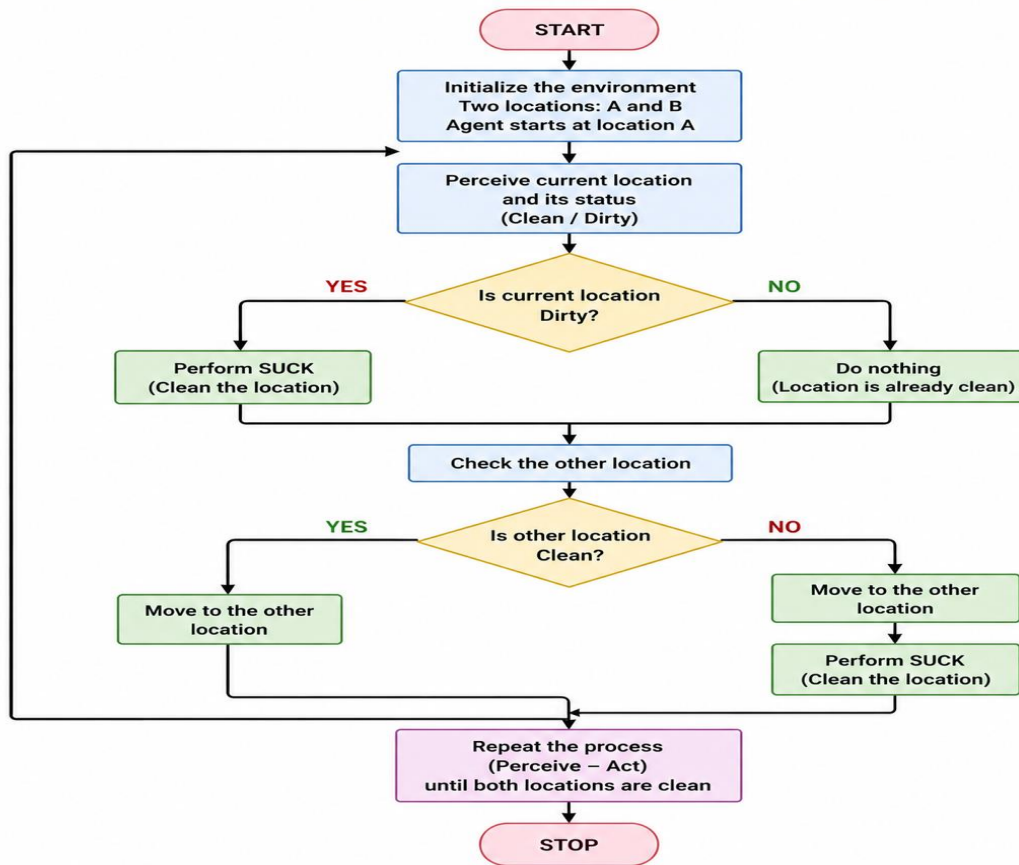
- To understand the concept of intelligent agents in Artificial Intelligence.
- To implement a simple goal-based vacuum cleaner agent.
- To learn how an agent perceives the environment and performs actions.

Algorithm

1. Start.
2. Define the environment with two rooms (A and B).
3. Initialize the rooms as dirty.
4. Place the vacuum cleaner in the initial location.
5. Check the current room condition.
6. If the room is dirty, clean the room.
7. If the room is clean, move to the next room.
8. Continue until all rooms become clean.
9. Display the final environment state.
10. Stop.

Flowchart:

VACUUM CLEANER PROBLEM – FLOWCHART



Code (Python)

```
def vacuum_cleaner():
```

```
    rooms = {  
        "A": "Dirty",  
        "B": "Dirty"  
    }
```

```
    location = "A"
```

```
    print("Initial State:")
```

```
    print(rooms)
```

```
while "Dirty" in rooms.values():

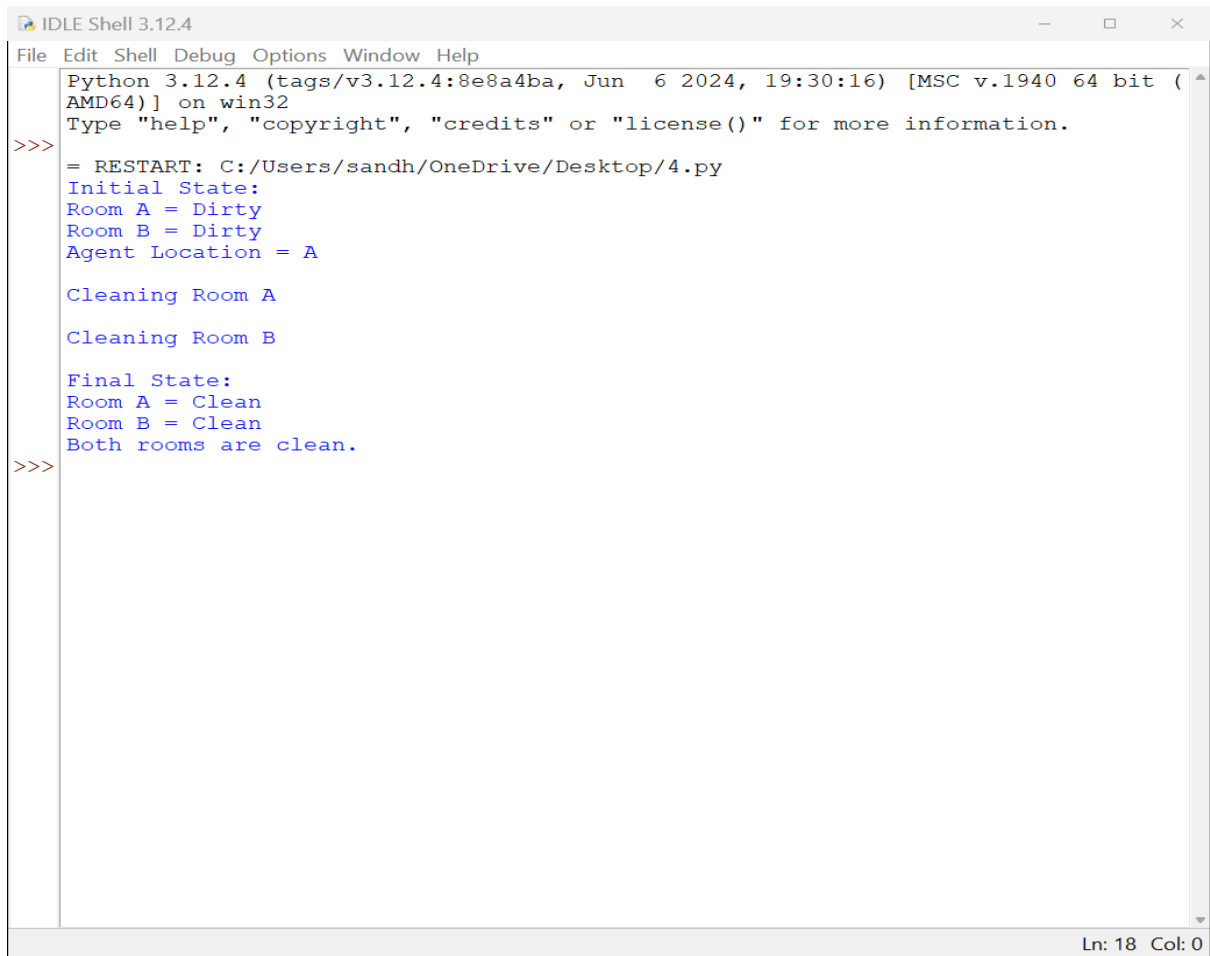
    if rooms[location] == "Dirty":
        print("Cleaning Room", location)
        rooms[location] = "Clean"

    else:
        if location == "A":
            location = "B"
            print("Moving to Room B")
        else:
            location = "A"
            print("Moving to Room A")

print("\nFinal State:")
print(rooms)
print("All rooms are clean!")
```

```
vacuum_cleaner()
```

Output



```
IDLE Shell 3.12.4
File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/sandh/OneDrive/Desktop/4.py
Initial State:
Room A = Dirty
Room B = Dirty
Agent Location = A

Cleaning Room A

Cleaning Room B

Final State:
Room A = Clean
Room B = Clean
Both rooms are clean.
>>>
```

Ln: 18 Col: 0

Result

Thus, the Python program for solving the **Vacuum Cleaner Problem** was executed successfully, and the vacuum agent cleaned all the rooms.

Conclusion

The experiment helped in understanding intelligent agents and their decision-making process in Artificial Intelligence. The vacuum cleaner agent successfully perceived the environment and performed actions to achieve the goal state.